PROJECT REPORT

Health360: Smart Healthcare & Risk Factor Analysis

Tableau-Based Healthcare Data Visualization Project

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1. INTRODUCTION

1.1 Project Overview

Health360: Smart Healthcare & Risk Factor Analysis is a healthcare data exploration and visualization project built using Tableau. It brings demographic, clinical and lifestyle indicators together to help users examine patterns in health conditions and associated risk factors. The deliverables include an interactive consolidated dashboard and a seven-part Tableau Story for guided exploration.

1.2 Purpose

- Make complex healthcare data easier to interpret using interactive visualizations.
- Explore patterns across age, BMI, blood pressure, glucose, cholesterol and lifestyle-related indicators where available in the dataset.
- Support exploratory comparisons and awareness of possible risk associations without making individual medical diagnoses.
- Present findings in a shareable dashboard and structured Tableau Story.

2. IDEATION PHASE

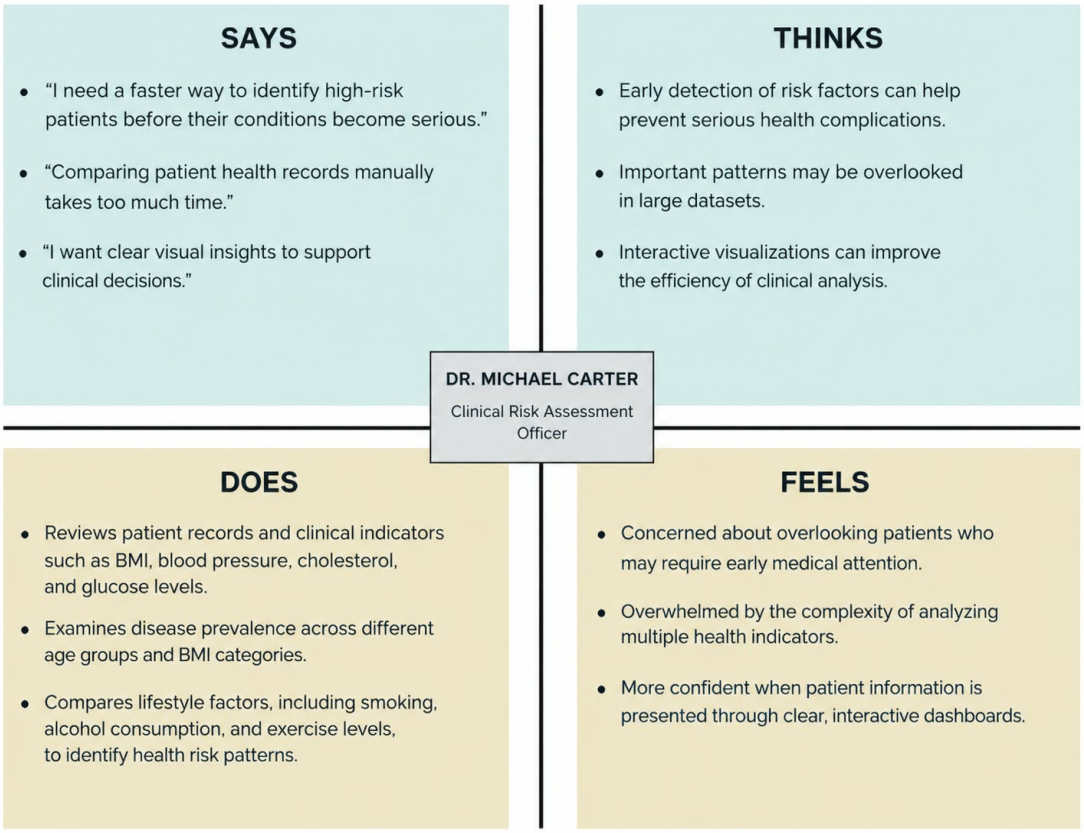
2.1 Problem Statement

Healthcare datasets contain numerous variables, making manual comparison of risk indicators time-consuming and difficult. Users need an accessible way to explore health trends, compare population groups and recognize patterns across clinical and lifestyle factors. Health360 addresses this analytical and communication gap through structured, interactive visualizations.

2.2 Empathy Map Canvas

Target user: a healthcare analyst or clinical risk assessment officer seeking understandable summaries of patient-level trends.

SAYS	THINKS
“I need clear views of health risk indicators.”	“Important patterns may be missed in large datasets.”
“Manual comparison takes too much time.”	“Interactive views could support faster exploration.”
DOES	FEELS
Reviews BMI, glucose, blood pressure and other available indicators.	Overwhelmed by scattered data and repeated comparisons.
Compares demographic groups and lifestyle categories.	More confident when findings are presented clearly.



2.3 Brainstorming

Ideas considered included static charts, summary tables, an interactive dashboard and a guided data story. The selected approach combines a Tableau dashboard for flexible exploration with a seven-section Tableau Story for step-by-step communication of insights.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey map

Stage	User action	Challenge	Health360 response
Discover	Seeks healthcare trends	Data is scattered	Single visualization entry point
Explore	Filters indicators and groups	Hard to compare	Interactive charts and filters
Interpret	Reviews patterns	Complex relationships	Organized dashboard and story
Share	Presents findings	Static reports are limiting	Published Tableau views

3.2 Solution Requirement

- Import a suitable healthcare dataset and verify field types and consistency.
- Provide views for relevant demographics, health measurements and lifestyle indicators present in the source.

- Offer interactive filters and comparative charts.
- Combine key views in one consolidated dashboard.
- Organize findings in a seven-section Tableau Story and publish the visualization.
- Keep the presentation understandable and avoid implying causal or diagnostic conclusions.

3.3 Data Flow Diagram

Step	Data flow
1. Input	Healthcare dataset
2. Preparation	Inspect fields, resolve quality issues as applicable, organize dimensions and measures
3. Analysis	Create Tableau worksheets and calculations as needed
4. Presentation	Consolidated dashboard + seven-section Tableau Story
5. Output	Interactive visual exploration and communicated insights

3.4 Technology Stack

Component	Technology / use
Visualization and analysis	Tableau Public
Source data	Healthcare dataset (exact source to be inserted)
Publishing	Tableau Public online
Presentation	Dashboard, Tableau Story and project report
Optional project webpage	HTML/CSS/JavaScript (only if included in submitted project)

4. PROJECT DESIGN

4.1 Problem Solution Fit

Target users such as students, researchers and healthcare analysts need clear, accessible views of complex healthcare information. Health360 addresses this need by converting raw records into interactive charts, comparisons and a guided Tableau Story. Its value lies in easier exploration and communication of population-level patterns, rather than automated clinical prediction.



4.2 Proposed Solution

The proposed solution is a Tableau-based healthcare analysis experience consisting of worksheets, one consolidated interactive dashboard and seven Story sections. Users can examine distributions, compare groups and review relevant risk-factor patterns, subject to the fields and limitations of the underlying dataset.

4.3 Solution Architecture

Healthcare dataset → data inspection and preparation → Tableau data model and worksheets → consolidated dashboard and seven-part Story → Tableau Public publication → viewer exploration.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work is organized into short sprints. Story points are relative effort estimates for documentation and planning purposes.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members

Sprint-1	Dataset acquisition	USN-1	As an analyst, I can load healthcare data for analysis.	2	High	—
Sprint-1	Data preparation	USN-2	As an analyst, I can inspect fields and handle identified quality issues.	3	High	—
Sprint-1	Visualization	USN-3	As a viewer, I can see charts of key health indicators.	5	High	—
Sprint-2	Dashboard	USN-4	As a viewer, I can compare indicators through an interactive dashboard.	5	High	—
Sprint-2	Story	USN-5	As a viewer, I can follow a seven-section narrative of insights.	3	High	—
Sprint-2	Publishing	USN-6	As a viewer, I can open the published Tableau project.	2	Medium	—
Sprint-3	Testing	USN-7	As a user, I can navigate and filter the views correctly.	3	High	—
Sprint-3	Documentation	USN-8	As an evaluator, I can review screenshots, links and findings.	2	Medium	—

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

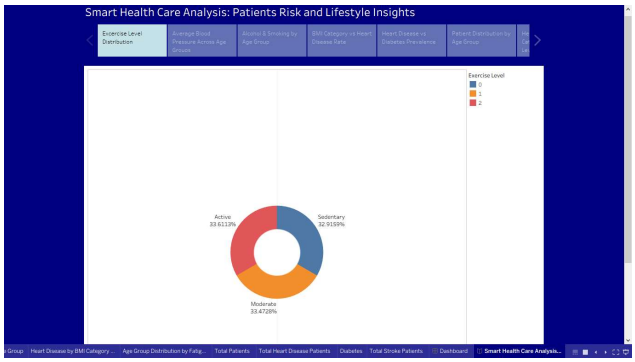
The following are suggested verification cases. Record actual observations after testing; no performance measurements are assumed here.

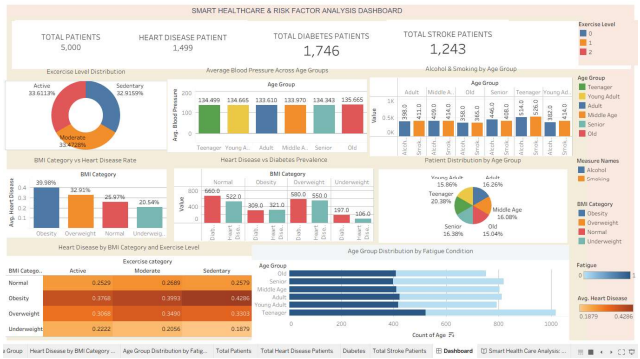
Test	Expected result	Observed result
Dataset refresh/load	Data loads without field errors	To be verified
Dashboard interaction	Filters update relevant views	To be verified
Story navigation	All seven story sections open	To be verified
Published access	Public link opens correctly	To be verified
Visual readability	Titles, legends and labels remain legible	To be verified
Response time	Interactions complete within acceptable time	Not measured

7. RESULTS

7.1 Output Screenshots

Insert actual screenshots from the completed Tableau workbook in the spaces below. Screenshots should match the submitted dashboard and story.





8. ADVANTAGES & DISADVANTAGES

Advantages

- Interactive exploration is more flexible than a static report.
- Dashboard and Story support both independent analysis and guided presentation.
- Visual comparisons can help users recognize patterns quickly.
- Tableau Public makes the published work easy to share.

Disadvantages

- Findings depend on the completeness and quality of the dataset.
- Observed associations do not establish causation or diagnose individuals.

- Public publishing is unsuitable for confidential or identifiable patient data.
- Complex views may require a learning period and an internet connection for published access.

9. CONCLUSION

Health360 demonstrates how Tableau can organize healthcare indicators into an accessible, interactive analytical experience. The consolidated dashboard and seven-section Story make it easier to explore patterns, compare groups and communicate observations. Interpretations remain limited to the available dataset and should not be treated as clinical advice.

10. FUTURE SCOPE

- Expand analysis with additional validated datasets and indicators.
- Improve accessibility, explanatory tooltips and mobile layout.
- Introduce deeper subgroup comparisons and time-based views where supported.
- Add documented data-quality checks and reproducible refresh procedures.
- Evaluate more advanced analytical models only after appropriate validation and privacy review.

11. APPENDIX

Dataset Link:

https://drive.google.com/file/d/1RDbCMrWJaLQUufYcGbI6m9LgTvuJQcv2/view

Project Demo Link:

Dashboard Link:-
https://public.tableau.com/views/HealthcareInsightsDashboard_17931122291220/Dashboard?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Story Link:-

https://public.tableau.com/views/HealthcareInsightsStory/SmartHealthCareAnalysisPatientsRiskandLifestyleInsights?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Demo Link:-

https://drive.google.com/file/d/1ypBpYvb2ijEwbjmrjHM8CoYIBplTE8--/view